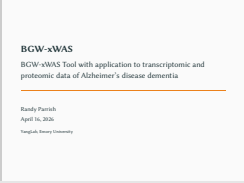
BGW-xWAS

BGW-xWAS Tool with application to transcriptomic and proteomic data of Alzheimer's disease dementia

Randy Parrish

April 16, 2026

YangLab, Emory University



The Yang Lab is interested in developing statistical and computational methods/tools for quantitative genomics data analysis and biomedical data analysis

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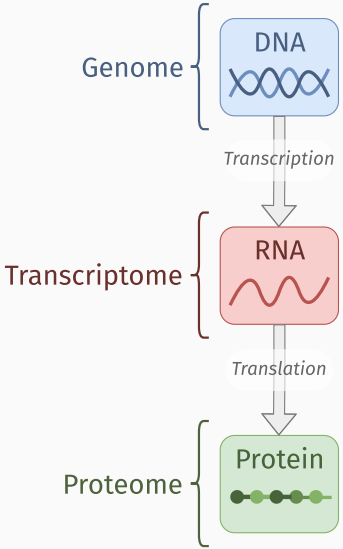
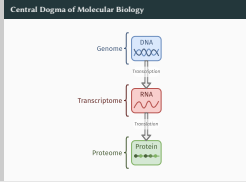
Central Dogma of Molecular Biology





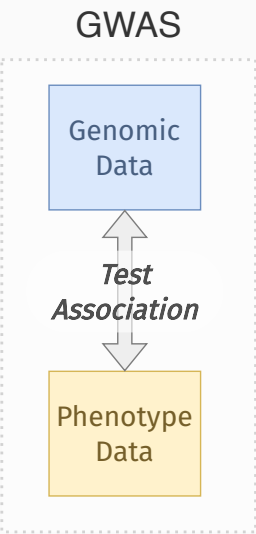
expression DNA code is turned into a functional product

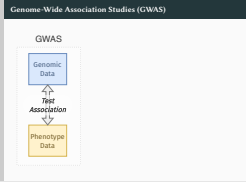
in typical datasets we have: 10mil variants

20k RNA transcripts

10k proteins

Genome-Wide Association Studies (GWAS)





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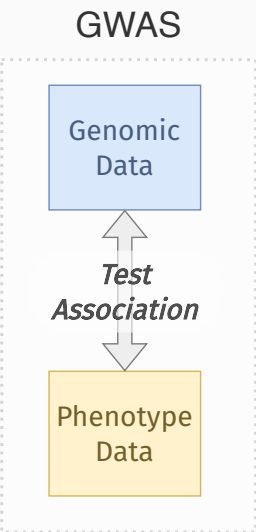

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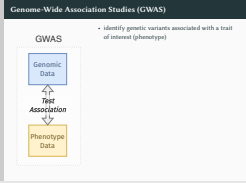

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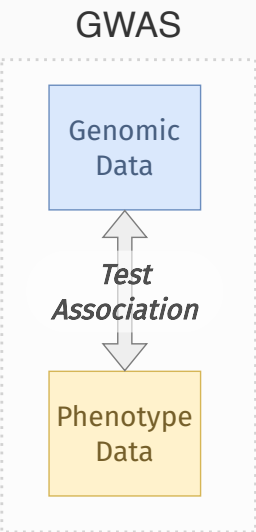

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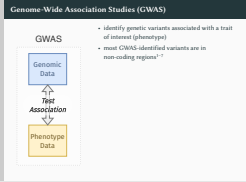

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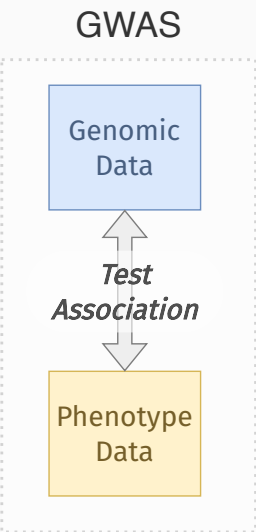

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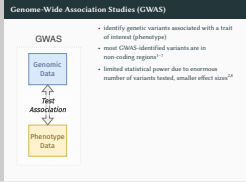

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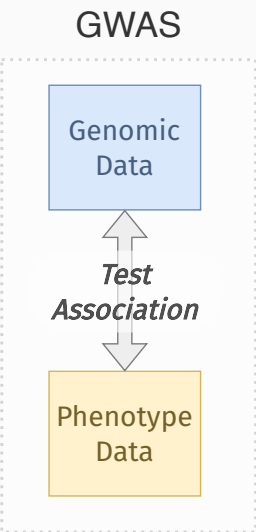

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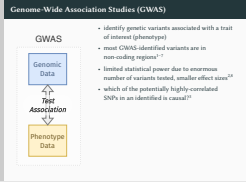

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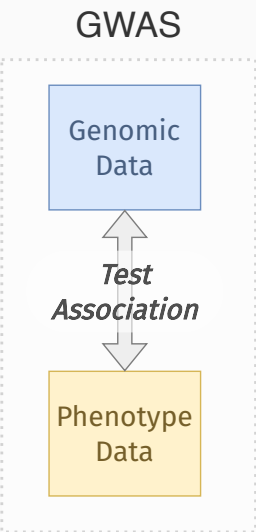

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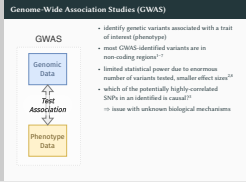

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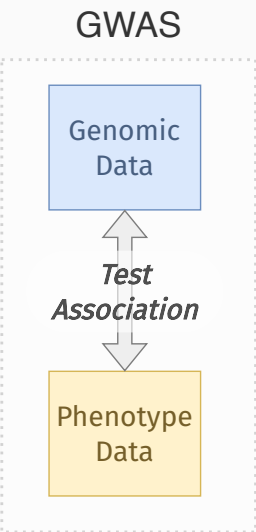

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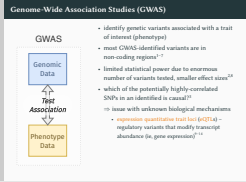

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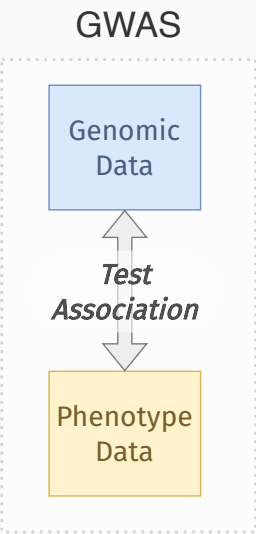

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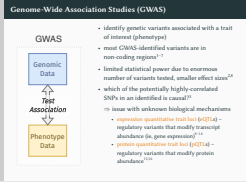
  - protein quantitative trait loci (pQTLs) – regulatory variants that modify protein abundance[15,16]



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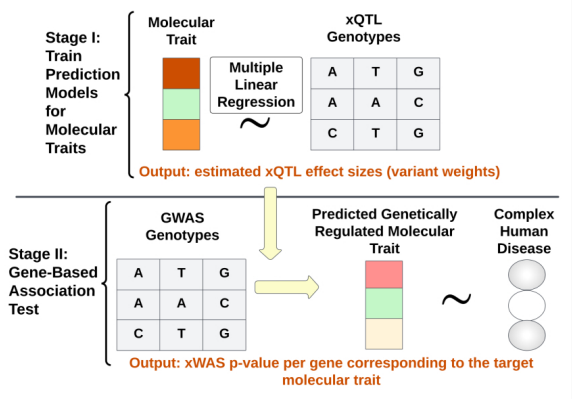
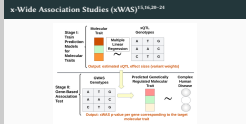
x-Wide Association Studies (xWAS)[15,16,20–24]

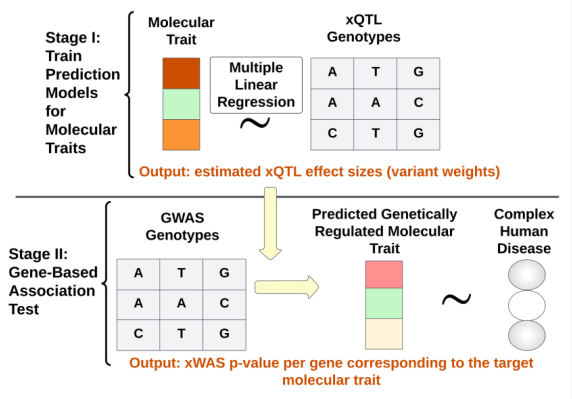




xWAS to find candidate causal genes in candidate gene analysis following genomewide association studies

studies ofter lack transcriptomic/proteomic data

x-Wide Association Studies (xWAS)[15,16,20–24]



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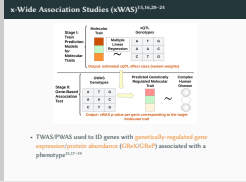

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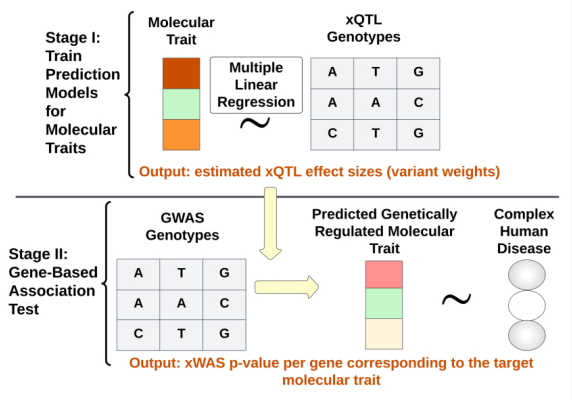

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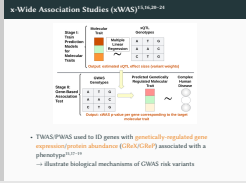
  → illustrate biological mechanisms of GWAS risk variants



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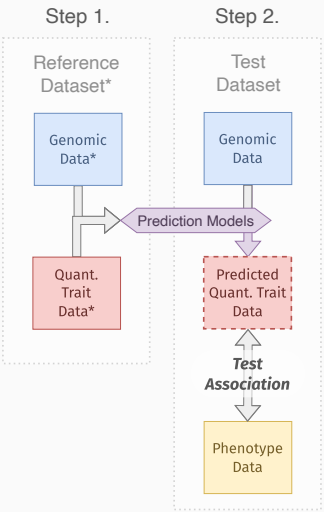

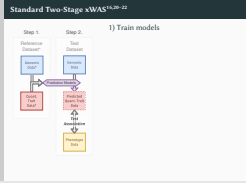

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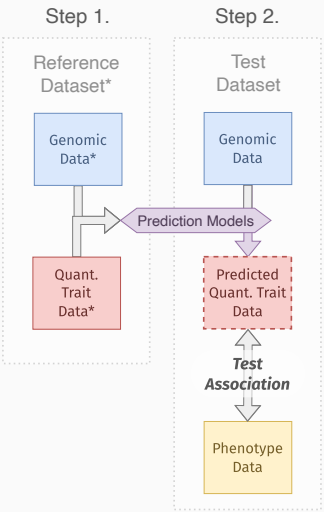

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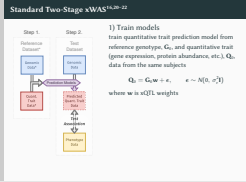
$$\mathbf{Q}_0 = \mathbf{G}_0\mathbf{w} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim N(0,\ \sigma_\epsilon^2\mathbf{I})$$

where $\mathbf{w}$ is xQTL weights



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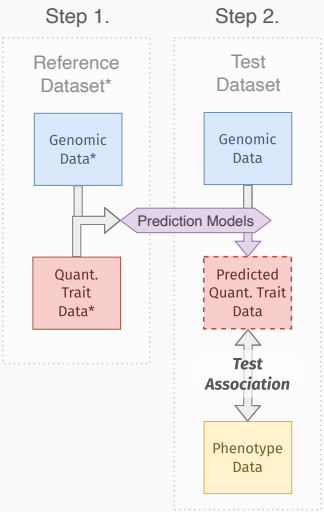

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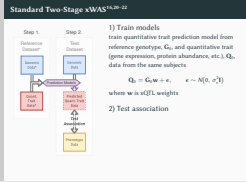

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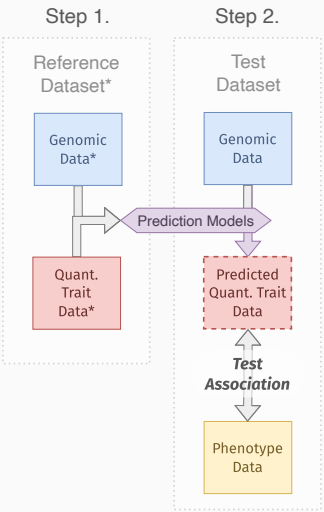

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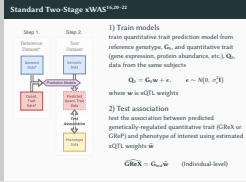
2) Test association

test the association between predicted genetically-regulated quantitative trait (GReX or GReP) and phenotype of interest using estimated xQTL weights $\widehat{\mathbf{w}}$

$$\widehat{\mathbf{GReX}} = \mathbf{G}_{\text{test}}\widehat{\mathbf{w}} \qquad \text{(Individual-level)}$$



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BGW-TWAS: Background

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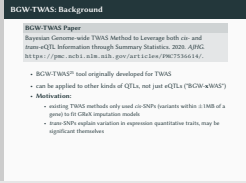
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  ⇒ including *trans*- variants expected to increase imputation accuracy of GReX and the power of TWASs

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BGW-TWAS[25] tool originally developed for TWAS
can be applied to other kinds of QTLs, not just eQTLs ("BGW-xWAS")
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generate single variant xQTL summary statistics (aka Score Statistics)

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Final Estimate: $(\widehat{\mathbf{w}}, \widehat{\mathbf{PP}})$ **Predicted GReX/GReP/etc.**:

$$\widehat{\mathbf{GReX}} = \sum_{i=1}^{p} \widetilde{G}_i(\widehat{PP}_i\, \widehat{w}_i) \qquad \widetilde{\mathbf{G}} = \text{test genotype data} \qquad \text{(Individual-level xWAS)}$$

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EM algorithm-iterative method to find (local) maximum likelihood or maximum a posteriori (MAP) estimates of parameters

MCMC algorithms-used to draw samples from a probability distribution, used to study complex, high dimensional probability distributions

iterate until maximum a posteriori (MAPs) estimates converge (MAP is the mode of the posterior density.)

BGW-TWAS: Summary-Level Association Test

$\widehat{PP}_i$ estimated posterior causal probability for SNP i
$\widehat{w}_i$ estimated xQTL effect-size for SNP i
$Z_{\text{GWAS},i}$ Z-score statistic values for single variant test (ie, GWAS)
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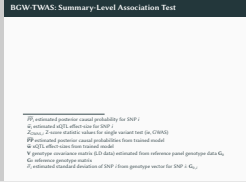

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Application TWAS & PWAS

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Outline

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Datasets

Results

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Training Data: ROS/MAP[26]

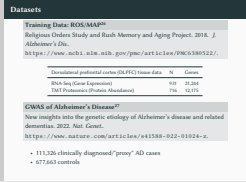
Religious Orders Study and Rush Memory and Aging Project. 2018. *J. Alzheimer's Dis.*.
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6380522/.

Dorsolateral prefrontal cortex (DLPFC) tissue data	N	Genes
RNA-Seq (Gene Expression)	931	21,244
TMT Proteomics (Protein Abundance)	716	12,175

GWAS of Alzheimer's Disease[27]

New insights into the genetic etiology of Alzheimer's disease and related dementias. 2022. *Nat. Genet.*.
https://www.nature.com/articles/s41588-022-01024-z.

- 111,326 clinically diagnosed/"proxy" AD cases
- 677,663 controls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TMT Tandem mass tags;
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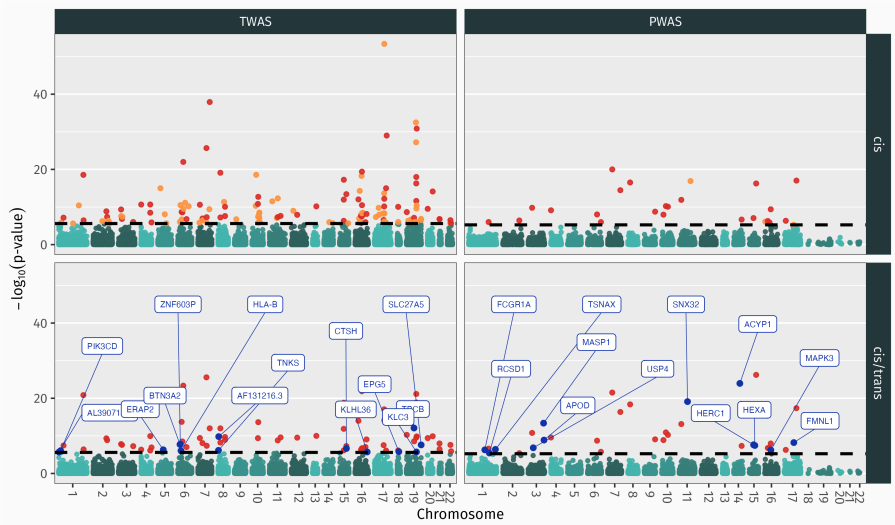
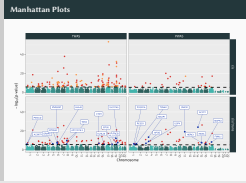
Manhattan Plots

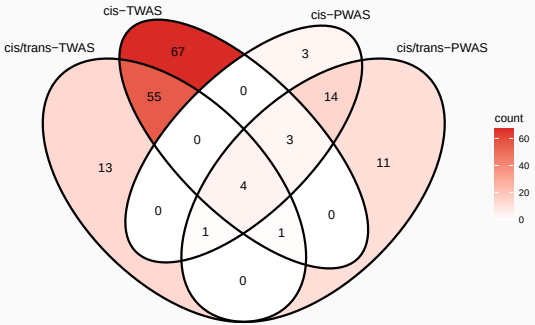




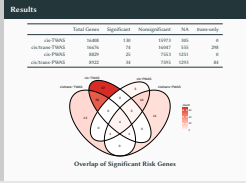
orange genes ID'd by only cis model
red genes ID'd both cis and cis/trans model
blue labeled genes are ID'd only by cis/trans model
EPG5 is hidden label

Results

	Total Genes	Significant	Nonsignificant	NA	*trans*-only
cis-TWAS	16408	130	15973	305	0
cis/trans-TWAS	16676	74	16047	555	298
cis-PWAS	8829	25	7553	1251	0
cis/trans-PWAS	8922	34	7595	1293	84



Overlap of Significant Risk Genes



	prop_sig	prop_nonsig	prop_NA
cis-TWAS	0.8%	97%	1.9%
cis/trans-TWAS	0.4%	96%	3.3%
cis-PWAS	0.3%	86%	14%
cis/trans-PWAS	0.4%	85%	14%

Results

Risk Genes Identified by all xWAS

CHR	Gene	Function/Disease Association
7	*EGFR*	epidermal growth factor receptor
8	*EPHX2*	lipid mediator; mutations associated with familial hypercholesterolemia
15	*LACTB*	regulator of mitochondrial lipid metabolism
17	*ACE*	angiotensin-converting enzyme

Gene Ontology Pathway Enrichment Analysis (pathDIP[28], PANTHER[29])

Enriched Paths Significant for all xWAS
Alzheimer disease-amyloid secretase
EGF receptor signaling
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all are previously ID'd GWAS risk genes

PANTHER pathways, ontology-based pathway database

* AD-amyloid secretase: disease pathway, enzymes that can cleave amyloid precursor protein

* EGF receptor signaling: cell signaling/proliferation; issues w/ EGFR cancer, inflammation, impaired wound healing

* Insulin/IGF ..: insulin receptor (IR), control cell fuel uptake/metabolism, defects insulin signaling involved in AD

* Interferon-gamma signaling: innate/adaptive immune system/intracell pathogens

* Ras Pathway: respond to cell proliferation cues, oncogene

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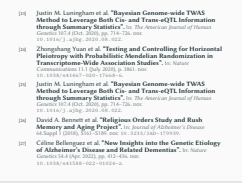

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